VISVESVARAYA TECHNOLOGICAL UNIVERSITY
Belagavi-590018

CAUVERY INSTITUTE OF TECHNOLOGY MANDYA-571401

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

MINI PROJECT PRESENTATION ON SMART HOME SYSTEM

Presented By:

Sangeetha T R (4CA23CS129)
Sanjana A N (4CA23CS130)
Spoorthi S (4CA23CS154)
Supreeth M C (4CA23CS161)

Under the Guidance of:

Prof. Amulya K S
Dept. of CSE
CIT Mandya

Contents:

Introduction:

- **Smart Home Parental Control System** that uses IoT and Blockchain to manage and secure device access for children.
- Children send **device access requests** which are recorded on the blockchain, ensuring the request cannot be altered or faked.
- Parents approve or deny these requests through their dashboard, and the decision is securely stored as a **tamper-proof blockchain transaction**.
- Approved requests automatically trigger the **IoT device to turn ON**, and Blockchain verification ensures only authorized access is granted.
- By combining **IoT automation** with **Blockchain security**, the system ensures trustworthy, transparent, and time-controlled smart device usage in homes.

Problem Statement:

- Unsupervised device usage and Lack of parental control
- Security vulnerabilities and Need for transparent authorization
- IoT-based access control
- Blockchain-based security

Objectives:

- Secure smart home access and Blockchain transparency
- IoT device automation
- Time-based device control
- User-friendly parent–child interface

Literature Survey:

Author & Year	Title / Work	Key Findings (Keywords)	Technology Used
Sharma et al., 2020	IoT-Based Home Automation System	Remote control, automation, sensor-based operation	IoT, Sensors, Wi-Fi, Microcontroller
Kumar & Singh, 2021	Parental Monitoring in Smart Devices	Usage tracking, alerts, limited control	Mobile App, Cloud Storage
Li et al., 2022	Blockchain for Secure IoT Access Control	Security, transparency, tamper-proof access	Blockchain, Smart Contracts, IoT

Existing System Disadvantages:

- No proper supervision
- Easy to bypass
- No time management
- Lack of security
- No transparency

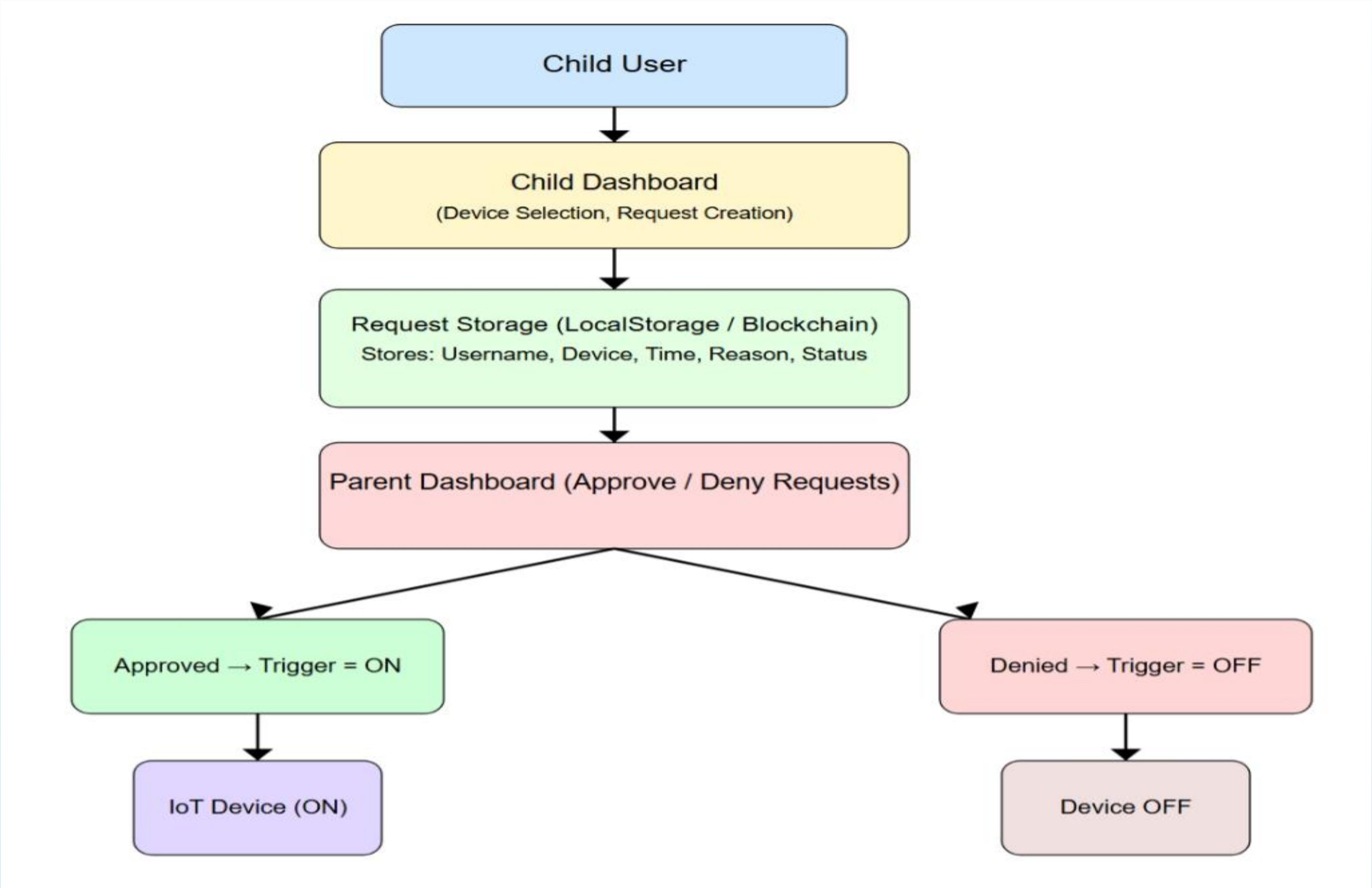
Hardware and Software Requirements:

- **Hardware:**
- ESP32 / NodeMCU
- Relay Module
- Power Supply
- Smart Device Control
- Wi-Fi Router
- **Software:**
- HTML / CSS / JavaScript
- LocalStorage / SessionStorage
- Blockchain (Solidity, Web3/Ethers)
- Arduino IDE
- MQTT Broker
- Web Browser

Proposed System:

- Blockchain Security
- IoT Device Control
- Parent Approval System
- Time-Based Access
- Smart Home Automation

Block Diagram:



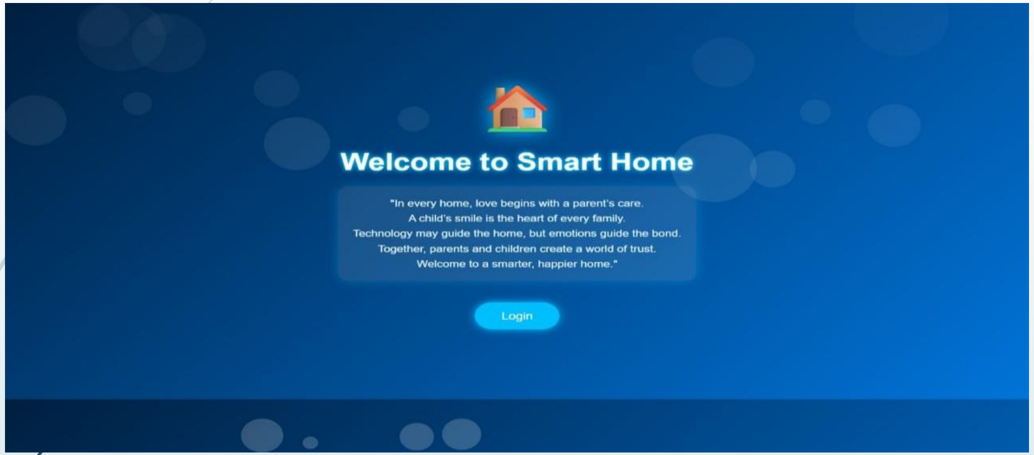
Work Flow:

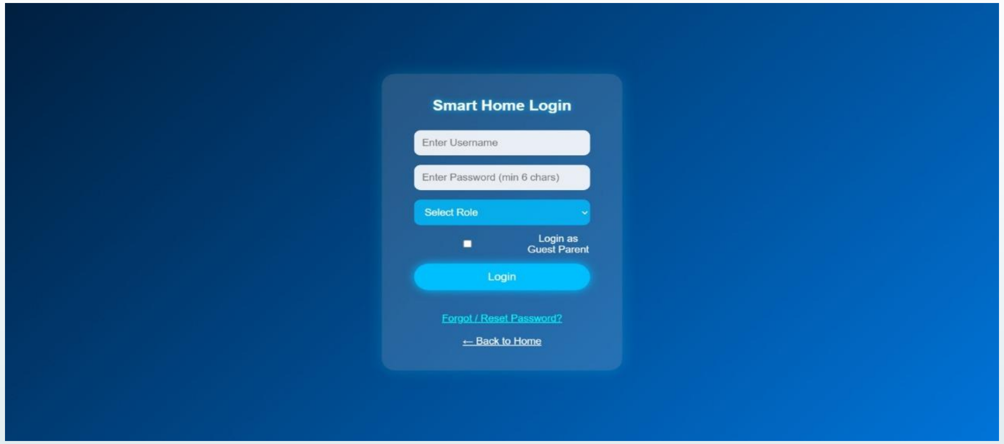
- 1. Child Request
- 2. Blockchain Storage
- 3. Parent Decision
- 4. IOT Device Action
- 5. Auto Expire

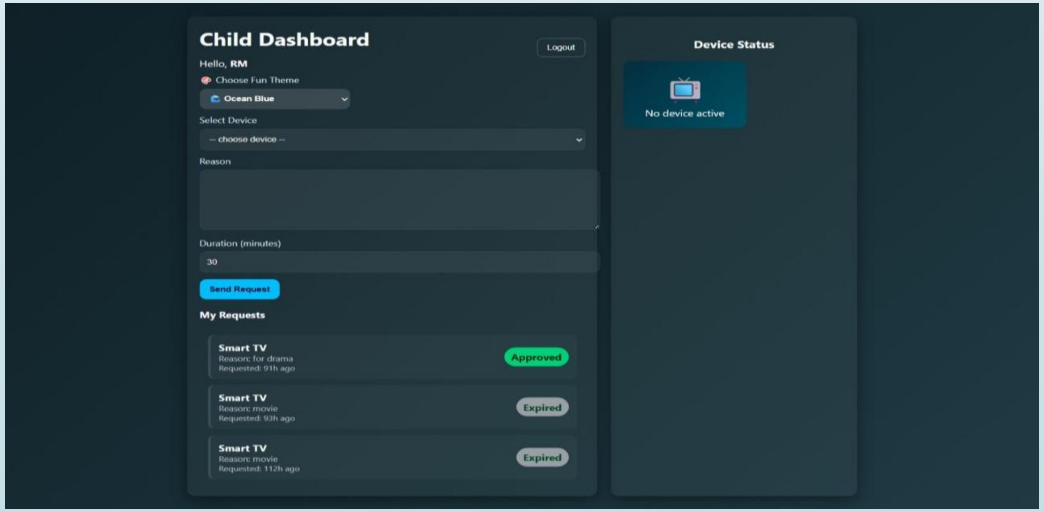
Methodology/Algorithms:

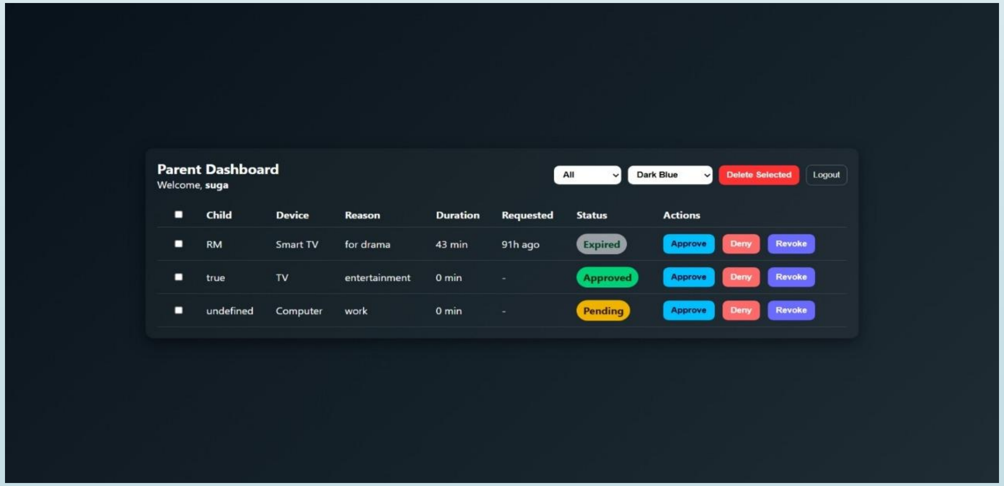
- User Authentication
- Child Request Creation
- Parent Approve/Deny Logic
- Smart Device Trigger (ON/OFF)
- Blockchain Verification
- Auto-Expiry Timer

Snapshots or Results:









Conclusion:

- The system helps parents safely control their children's smart device usage.
- IoT automatically turns devices ON/OFF based on approval.
- Blockchain keeps all actions secure and tamper-proof.
- It makes smart homes safer, transparent, and more reliable.

Future Work:

- Mobile App Integration
- Voice Assistant Support
- AI/ML Usage Analysis
- Multi-Device Expansion
- Cloud-Based Monitoring

References:

- 1. Sharma, P. (2020). IoT-Based Home Automation System. International Journal of Computer Applications.
- 2. Kumar, S. (2021). Parental Monitoring in Smart Devices. Journal of Emerging Technologies.
- 3. Li, X. (2022). Blockchain for Secure IoT Access Control. IEEE Internet of Things Journal.

Thank You